Replicating the Conservative Formula

Yingni Chang and Yikai Shen · MTH 9897 Project 8 · CRSP replication, 1929–2024

Paper: Blitz and van Vliet (2018), The Conservative Formula, Journal of Portfolio Management 44(7)

Replicate the US results on the paper's sample, 1929–2016

Then examine factor exposure, later periods and trading costs

Main recent comparison: 1990–2024; the paper ends in 2016

US REPLICATION

1929-2016

RECENT EXTENSION

2017-2024

What we test

01 REPLICATE

Can we reproduce the paper?

1929–2016
Return, volatility and turnover

→

02 EXPLAIN

How much return comes from known factors?

Carhart: primary model
FF5 + momentum: robustness

→

03 TEST RECENT PERFORMANCE

Does it still perform after trading costs?

Recent periods through 2024
10 / 30 bps and break-even costs
2017–2024: 0–30 bps grid

Time checks: 1929–1962 | 1963–1989 | 1990–2016 | 2017–2024; rolling 120-month windows

Uncertainty: Newey–West (12 lags) or block-bootstrap confidence intervals

The data

Monthly CRSP via WRDS: 1925-12 to 2024-12; extracted 2026-10-13.

Data item	Coverage / count
Security-month observations	3,954,591
Securities	26,746
Delistings	22,869 total; 2,530 imputed; 317 set to zero
Eligible universe, December 1928	393
First full 1,000-stock universe	1964-06

01 Keep delisted stocks

Include terminal returns to reduce survivorship bias.

02 One common share class

Keep one class per company each month to avoid double counting.

03 Respect early coverage

393 eligible stocks at 1928-12; the universe first reaches 1,000 in 1964-06.

One share class per company; dividends implied from total minus price return. Missing held returns: stale-price carry or in-gap price ratio; every case counted. WRDS provides no release identifier; time stamp and file hashes identify the extract.

The portfolio rule

1,000 largest eligible US stocks

36-month history screen, then market capitalization

↓

500 lower-volatility stocks

$$\sigma_{i,t} = \sqrt{12} \times \mathrm{Std}(R_{i,t-35}, \ldots, R_{i,t})$$

↓

Momentum

$$MOM_{i,t} = \prod_{j=1}^{11}(1 + r^{\text{price}}_{i,t-j}) - 1$$

Net payout yield

Dividends + net share repurchases

Average the two ranks; select the best 100

↓

100 stocks

Equal-weighted portfolio; rebalance quarterly

Convention	Implementation
Share classes	Largest common class per company each month; share codes 10/11.
Size measure	Security market capitalization after the history screen.
Dividends	Implied from RET - RETX; legacy RETX includes non-ordinary distributions.
Missing signals	Partition by volatility first, then rank complete cases.
Early universe	Use all eligible names if fewer than 1,000: 393 at 1928-12; full 500-name pool from 1964-06.
Missing returns	Stale carry (0) or an in-gap price ratio.

Weights drift between rebalances; delisted holdings become cash; costs charged per dollar traded

Controls: Speculative (high-volatility half, weakest scores), standalone signals, Small, CRSP VW market

Excluded: Value (no book equity in CRSP) and Table 2 panels C–D (factors not in our sources)

Building the signals

Net payout yield = dividend yield + net share repurchase proxy

$$\mathrm{NPY}_{i,t} = \frac{\sum_{j=0}^{11} D_{i,t-j}^{\mathrm{adj}}}{P_{i,t}^{\mathrm{adj}}} + 1 - \frac{S_{i,t}^{\mathrm{adj}}}{\frac{1}{24}\sum_{j=0}^{23} S_{i,t-j}^{\mathrm{adj}}}$$

D = adjusted cash dividends; P = adjusted price; S = adjusted shares. Both windows include month t.

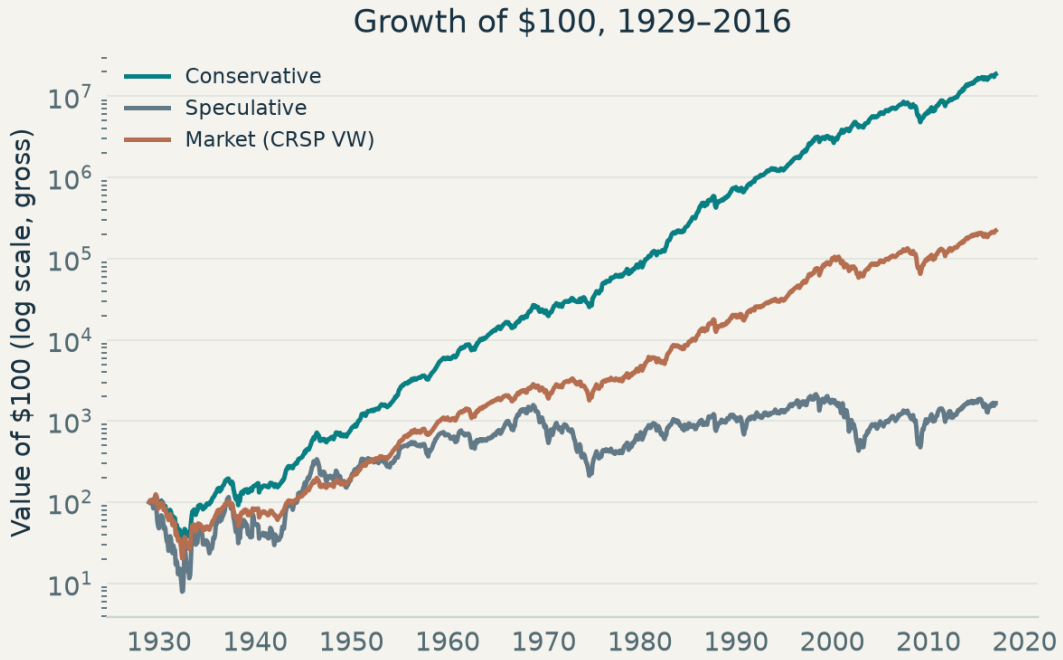
Signal	History	Calculation
Volatility	36 months	Annualized standard deviation of consecutive monthly total returns
Momentum	11 months	Compound price returns; exclude the most recent month
Dividend yield	12 months	Trailing split-adjusted dividends / current adjusted price
Net payout yield	24-month share average	Dividend yield + 1 − latest adjusted shares / average adjusted shares

A falling share count raises payout yield; new share issuance lowers it.

Quarter-end information determines next month's holdings. No forward return enters selection.

Our results versus the paper

Metric	Ours	Paper
Conservative CAGR	14.80%	15.1%
Conservative volatility	17.06%	16.5%
Market CAGR	9.15%	9.3%
Speculative CAGR	3.24%	—
Speculative volatility	35.9%	—
Quarterly turnover	38.9%	32%
Decades beating market	9 / 10	—



Gross, 1929–2016. Turnover gap unexplained; paper measure not stated. — = not shown.

paper / ours	Formula	Market	Small	Momentum	Lowvol	NPY
Return (compounded)	15.1% / 14.8%	9.3% / 9.1%	12.0% / 13.0%	15.9% / 15.5%	10.2% / 10.4%	14.5% / 14.2%
Volatility	16.5% / 17.1%	18.7% / 18.7%	31.4% / 33.5%	25.7% / 26.1%	13.3% / 14.4%	20.1% / 21.1%

Factor-adjusted returns

Annual alpha; gross unless labeled net. CMS = Conservative − Speculative; net CMS charges both legs.

Portfolio	Period	Model	Ours α	Paper α	Our t
CMS	1963–2016	CAPM	14.20%	14.0%	—
CMS	1963–2016	FF5	8.69%	8.8%	—
CMS	1963–2016	FF5 + MOM	3.12%	3.3%	—
CMS	1929–2016	CAPM	12.33%	13.9%	—
CMS	1929–2016	Carhart	7.74%	8.8%	—
Conservative	1929–2016	Carhart	3.23%	—	4.63
Conservative	1963–2016	FF5 + MOM	−0.11%	—	−0.14
CMS, net 30 bps	1929–2016	Carhart	5.78%	—	3.99

Full-sample gap remains. Adding profitability and investment reduces long-only alpha. — means the comparison or t-statistic is not shown.

How costs change the result

Annual cost estimate: quarterly turnover × 8 × cost.

1929–2016 measure	Value
Quarterly turnover: ours	38.9%
Quarterly turnover: paper	32%
Net CAGR at 30 bps	13.73%

Break-even cost (bps per dollar traded)

Period	Carhart alpha	Excess return
1929–1962	162	178
1963–1989	66	187
1990–2016	52	84
1990–2024	36	52
2017–2024	none	none

2017–2024: gross excess return and alpha already ≤ 0, so no break-even.

Early-period break-evens are high, but so were real trading costs then.

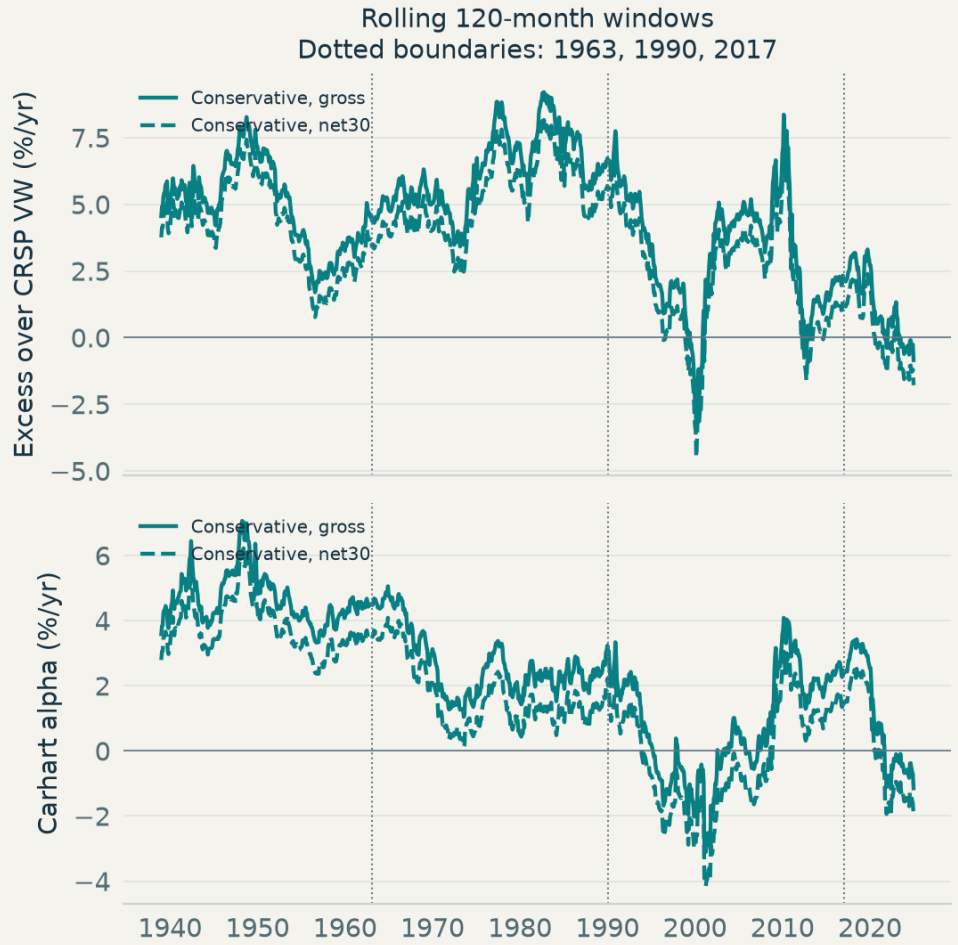
Results by period

Carhart alpha (gross): 4.7% → 2.1% → 1.7% → −1.2%

Carhart alpha is statistically significant in the two earlier periods

2017–2024 is only 96 months: interval [−5.1%, 2.7%]

period	months	excess, gross	Carhart α gross (t)	Carhart α net 30 (t)
1929–1962	408	5.2%	4.7% (5.03)	3.8% (4.14)
1963–1989	324	6.4%	2.1% (2.90)	1.2% (1.56)
1990–2016	324	2.7%	1.7% (1.34)	0.7% (0.56)
2017–2024	96	−1.7%	−1.2% (−0.61)	−2.1% (−1.06)
1990–2024	420	1.7%	1.2% (1.05)	0.2% (0.18)



The recent sample: 1990–2024

Portfolio / measure	Gross	Net 10 bps	Market
Conservative: excess over market	1.69% [−2.0%, 5.4%]	1.37% [−2.3%, 5.1%]	—
Conservative: Carhart alpha	1.17% (t 1.05)	0.85% (t 0.76)	—
Conservative: FF5 + MOM alpha	−1.03% (t −1.21)	−1.36% (t −1.59)	—
Conservative: CAGR	12.2%	11.9%	10.1%
Conservative: volatility	13.5%	13.5%	15.2%
CMS: Carhart alpha	5.78% (t 3.45)	5.10% (t 3.04)	—

Carhart alpha across cost levels; 30 bps is the paper's high-cost case

1990–2024	0 bps	5 bps	10 bps	30 bps
Carhart alpha	1.17% (t 1.05)	1.01% (t 0.91)	0.85% (t 0.76)	0.20% (t 0.18)

10 bps is the paper's low-cost case; quoted half-spreads of the top 1,000 stocks averaged 2.4 bps in 2016–2024. Brackets are 95% confidence intervals; returns and alphas are annualized. CMS charges costs on both legs and omits borrow and financing costs.

Dividends and share repurchases

NPY = trailing dividend yield + 1 − latest adjusted shares / 24-month average adjusted shares.

Repurchases raise NPY; issuance lowers it. Splits must leave adjusted shares unchanged.

Annual arithmetic return, gross	1929–2016	1990–2024
Net payout rule	15.33%	12.53%
− Dividends only	14.11%	10.90%
= Gain from the share-count term	1.23%	1.63%

Same universe, low-volatility pool, momentum signal, weights and quarterly dates; only the payout ranking differs. Share changes measure net repurchases after issuance, not buyback cash flow. Differences use unrounded values.

Industry allocation and stock selection

Reference: cap-weighted formation universe, quarterly constituents fixed, weights drift.

Excess over CRSP market, gross	1929–2016	1990–2024
Industry allocation	0.82%	−0.04%
+ Stock selection	3.80%	1.08%
+ Reference universe − CRSP market	0.18%	0.66%
= Excess over CRSP market	4.80%	1.69%

Industry quota variant against the baseline

Annual arithmetic return, gross	1929–2016	1990–2024
Industry quota variant	15.18%	12.71%
− Baseline (global screen)	15.33%	12.53%
= Difference	−0.15%	0.19%

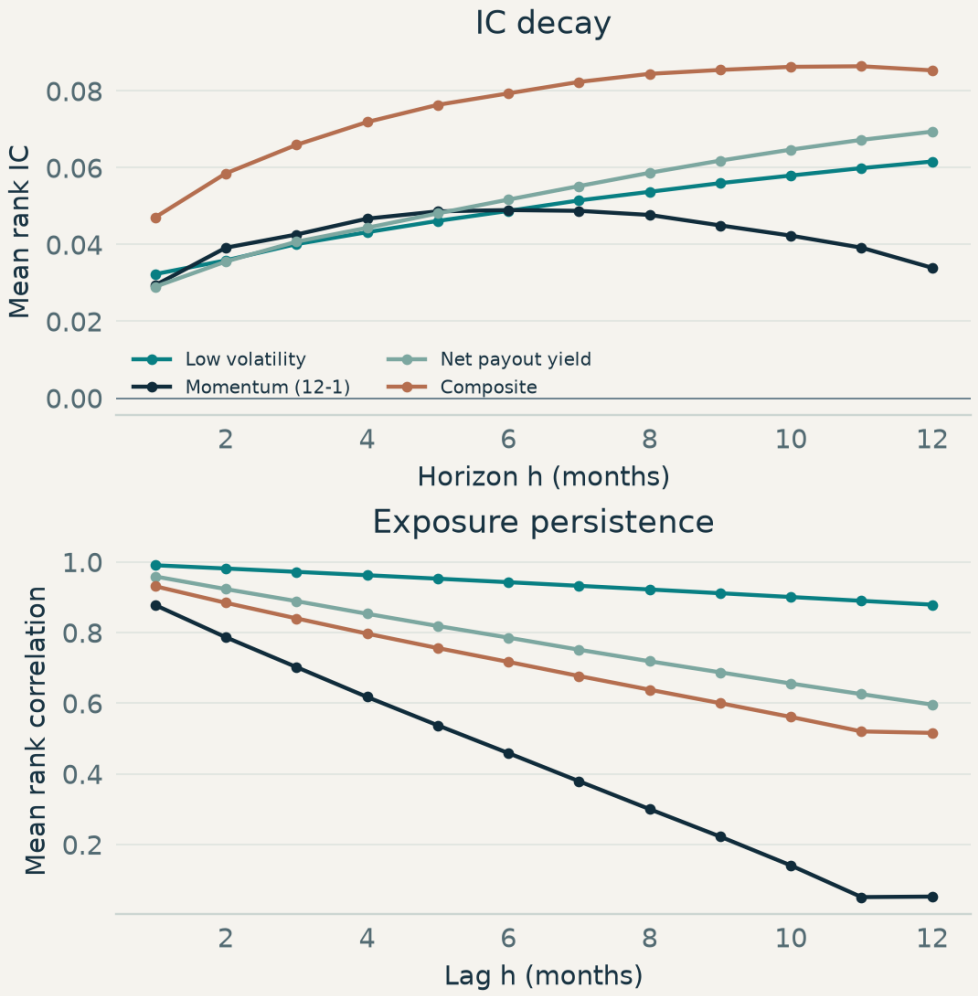
Sums and differences use unrounded values. Quota: 100 slots by FF12 security counts, rule applied within each industry; exploratory, and contributions are not causal.

Signal tests

Diagnostic	Result
Composite rank IC, 1 month	0.047 (t 8.8)
D10−D1, 1963–1989	1.02%
D10−D1, 1990–2016	0.68%
D10−D1, 2017–2024	0.40%
FMP information ratio, K = 1	0.59
FMP information ratio, K = 12	0.29
FMP IR, K = 1, 2017–2024	0.29

D10−D1 is a monthly decile spread. FMP portfolios are gross diagnostics and may have leverage.

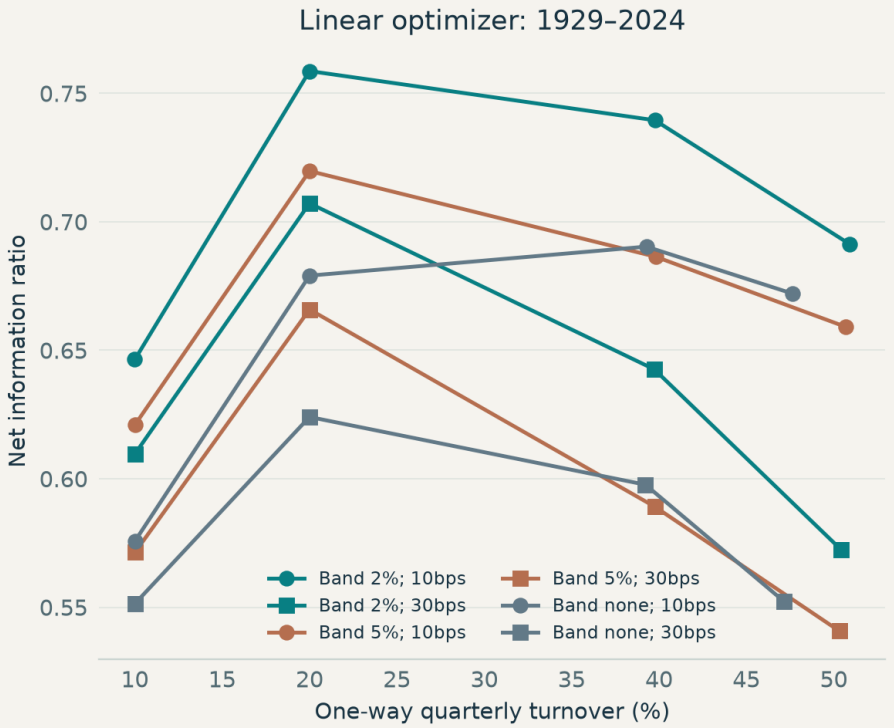
Low volatility: positive rank IC, flat mean spread (right-skewed high-volatility deciles).



Long/short and linear optimization

Long/short: n = 3, 10 bps and borrow. Linear alpha: net 30 bps, annualized.

Trading / optimization measure	Result
Long/short annual excess over RF	0.64%
Long/short information ratio	0.03
Long/short max drawdown	−99.97%
2020s capacity, scaled to 2024	$7.9B
Linear alpha, 1990–2024	1.31% to 1.95%
95% intervals excluding zero	2 / 4
Linear alpha, 2017–2024	−2.71% to 0.07%
95% intervals excluding zero	0 / 12



Monthly reweighting was chosen after buy-and-hold losses (1932-08, 1933-05). Capacity assumes 10% of volume, scaled to 2024 market size; no impact model. Linear recent examples use 2–5% industry bands and 20–40% turnover budgets; intervals are unadjusted, selected from a 24-run grid. Return-scaled alpha is a later sensitivity. We also test a portfolio covariance penalty.

Mean-variance portfolios

Maximize expected return − half risk aversion × portfolio variance − trading costs

Covariance: trailing 36 monthly returns, 50% diagonal shrinkage; the window ends at formation

Return alpha: completed trailing IC × annual volatility × √(3/12) × composite exposure

Quarterly: 2% stock cap, industry bands ±5pp, two-way turnover budget 40%; scaled costs start at 30bps

Two fixed risk aversions (5 and 20); max constraint violation 4.9e-09; turnover relaxations 0

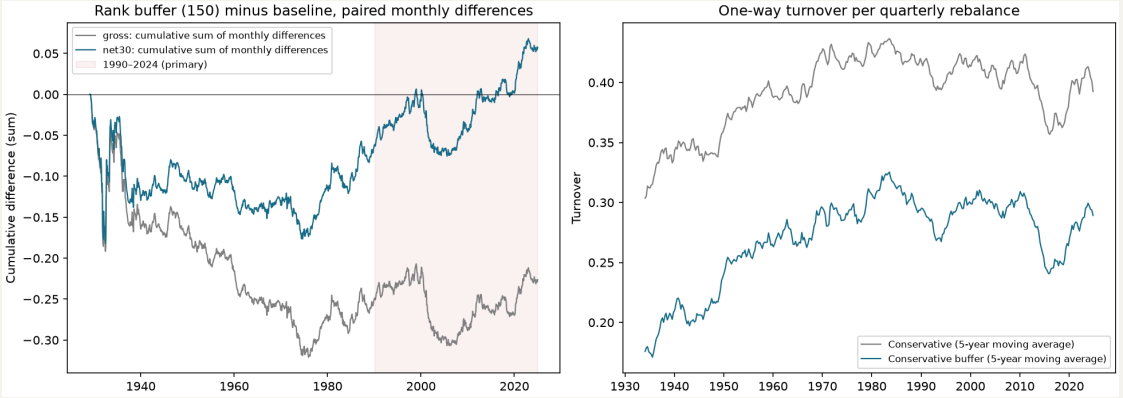
We chose these settings after examining the baseline; an unseen test is still needed

1990–2024	Net CAGR	Volatility	IR	Market corr.
γ = 5	11.59%	12.51%	0.14	0.89
γ = 20	10.39%	11.51%	−0.03	0.87

A buffer to reduce trading

Keep holdings while eligible and within the best 150 ranks.

1990–2024 comparison	Result
Baseline quarterly turnover	39.8%
Buffer quarterly turnover	28.4%
Relative turnover reduction	28.7%
Paired net-30 gain, per year	0.34% [−0.10%, 0.78%]
Paired gross gain, per year	0.06%



Net paired interval includes zero: not significant. The buffer changes holdings as well as costs.

The recent period was already seen when we tested this variant.

What we conclude

Return and volatility are close to the paper; turnover and some regression results differ

Much of the historical return is associated with known equity factors

Alphas decay; 1990–2024 net of 30 bps: alpha 0.20% [−2.0%, 2.4%]

The recent net-alpha interval includes zero

Lower volatility is the most consistent feature of the portfolio

Limitations: no CRSP release identifier; implied dividends; share-class and size conventions; turnover above the paper's; selection across grids; a short post-sample

2017-2024 trading-cost scenarios

Before costs

11.01%

Annual compound return

At 30 bps

9.97%

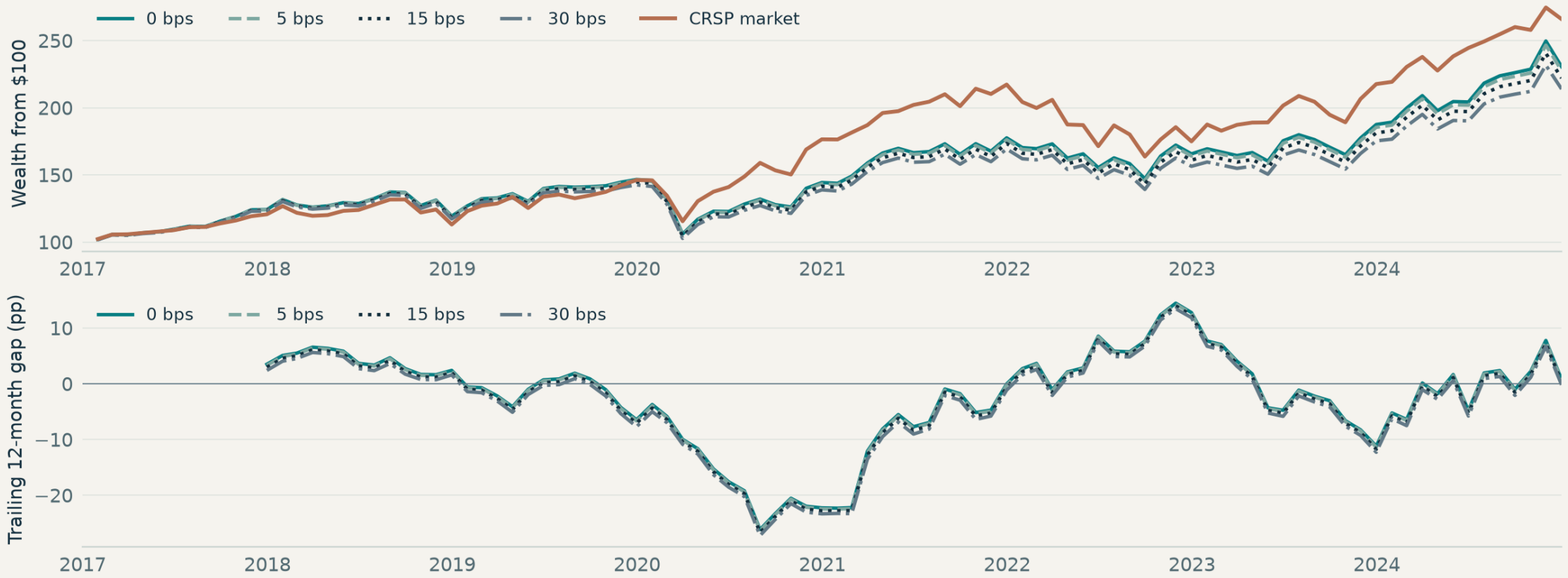
Market: 13.01% per year

Same holdings and trades at every cost level. Both purchases and sales pay the quoted bps.

Cost (bps)	Annual return	Above market / yr	Carhart alpha / yr	Final $100
0	11.01%	-1.74%	-1.21%	$230.55
5	10.83%	-1.90%	-1.36%	$227.69
10	10.66%	-2.06%	-1.51%	$224.86
15	10.49%	-2.22%	-1.66%	$222.08
20	10.32%	-2.38%	-1.80%	$219.32
25	10.14%	-2.54%	-1.95%	$216.60
30	9.97%	-2.70%	-2.10%	$213.91

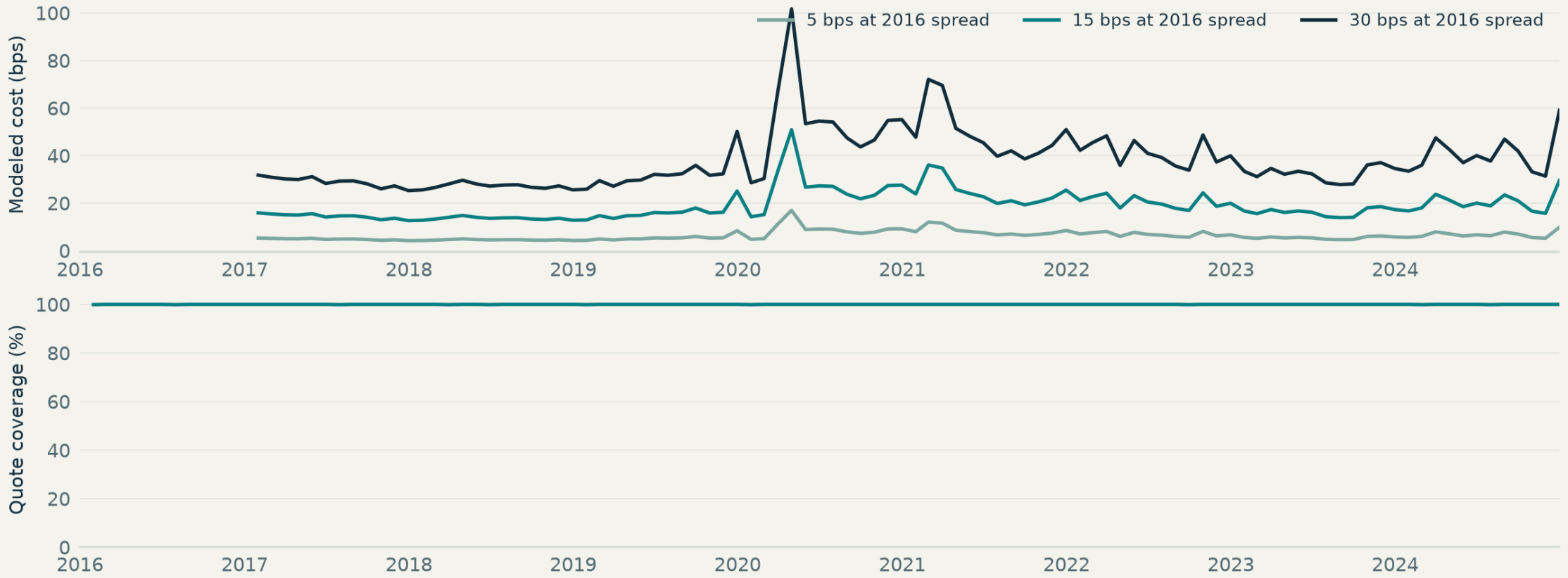
Annual return is compounded. Above-market return is 12 times the mean monthly strategy-minus-market return. Alpha uses the US Carhart model. All recent alpha intervals include zero.

When the recent return gap developed



Monthly wealth and trailing 12-month compounded return gaps. Rolling windows overlap and describe this sample. They do not establish a forecast or a structural break.

Costs that change with quoted spreads



Cost in month t = reference bps x previous-month average half spread / fixed 2016 average. Quote coverage ranges from 99.9% to 100.0%. Quotes proxy relative costs and omit impact.

UK data and implementation

Item	UK extension
Source	Compustat Global daily, 2013-2024. Last observation of each month.
Sample	UK-incorporated London ordinary GBP issues. Includes inactive securities.
Same portfolio rule	36-month volatility, 12-1 momentum, net payout, quarterly equal weights.
Universe and holdings	961-1000 eligible universe names, 100 holdings. A top-10% variant is separate.
Own-market comparator	Same eligible UK universe, cap weights reset quarterly, gross.
Interpretation	Current country/exchange headers, missing terminal payments and local currencies limit comparability.

Adjusted price = PRCCD / QUNIT / AJEXDI. Adjusted shares = CSHOC x AJEXDI. Total-return index = adjusted price x TRFD. No UK factor alpha is estimated.

UK and US after the paper period

Same 2017-2024 window and 100-stock rule. Compare each strategy with its own country benchmark.

2017-2024, net 10 bps	Annual return	Own benchmark	Above market / yr
US, USD	10.66%	13.01%	-2.06%
UK base, GBP	7.23%	5.40%	2.01%
UK exit stress, GBP	1.65%	3.55%	-1.58%
UK plus 50bps buy tax	6.34%	5.40%	1.17%

UK base sells disappearing positions at last known value. Exit stress loses all remaining value. The separate tax scenario adds 0.5% on purchases with no exemption assumed. UK figures are GBP, US figures USD. The UK advantage reverses under the exit stress. This is exploratory evidence.